

# IPCC Tier 2 Livestock GHG Uncertainty Calculator

Beta Testing — How to Review the Tool and Send Feedback

CIAT / CGIAR Alliance — ClimateActionNetZero

## Help us improve the tool — a 20-minute review

Thank you for agreeing to review the **IPCC Tier 2 Livestock GHG Uncertainty Calculator**. This short guide walks you through a quick hands-on test using a small practice dataset, then shows you how to record and send your feedback. Plan for about **20–30 minutes**. You don't need to be an R user or install anything — the tool runs in your web browser.

### What you'll need

Three files (attached to the invitation email, or downloadable from the app's **Resources** tab):

1. **beta\_sample\_inventory.xlsx** — a small practice dataset to load into the tool.
2. **beta\_feedback\_template.xlsx** — the form where you write your feedback.
3. **This guide**.

### Step 1 — Open the tool

Go to <https://mlolita26.shinyapps.io/cattle-ghg-uncertainty/>. The calculator opens straight away — **no login is needed**. (Only the optional *AI Translator* on the Resources tab asks for an email; you can ignore it for this review.)

### Step 2 — Load the practice dataset (*about 5 min*)

The practice file is a small, made-up inventory: one anonymised “**Country**” with **two production systems** (Commercial dairy and Extensive beef), each split into **two animal sub-categories**. It's deliberately tiny so you can see how every part of the tool behaves without wading through a real national dataset.

1. Open the **Data Input** tab.
2. Under **Upload**, select **beta\_sample\_inventory.xlsx**.
3. Tick the **emission sources** to include — start with all of them (enteric CH<sub>4</sub>, manure CH<sub>4</sub>, direct and indirect N<sub>2</sub>O).

*(The built-in “Country X / Country Y” examples in the dropdown are also there to play with, but please base your review on the practice file so we're all looking at the same thing.)*

### Step 3 — Walk through the tabs (*about 10 min*)

Go through these in order, and jot down anything that is **confusing, wrong, or missing**:

- **QA/QC** — the automatic data checks and any flags they raise.
- **Uncertainty** — the  $\pm$  ranges shown for each parameter.
- **Correlations** — how parameters move together (the practice file includes multi-year data, so this tab has something to display).
- **Simulate** — run the Monte Carlo simulation; look at the totals and the distribution chart.
- **Sensitivity** — which inputs drive most of the uncertainty (the tornado chart).
- **IPCC Report** — the summary tables and the downloadable Word report.

If you have a real inventory of your own, please try uploading that too and note how it goes.

### Step 4 — Record your feedback (*the main step*)

Open **beta\_feedback\_template.xlsx** and add **one row per observation** in the **Feedback** sheet. Use the drop-down menus in each cell:

- **Tab** — where you saw it (Home, Data Input, QA/QC, . . . , Contact).
- **Type** and **Severity** — see the tables below.
- **Description** — what you observed and what you expected instead.
- **Suggested fix / comment** — optional; your idea for the right behaviour.

Leave the **Status** column blank — the development team fills that in as items are addressed.

| Type               | Use it when...                                                               |
|--------------------|------------------------------------------------------------------------------|
| <b>Bug</b>         | The tool crashes, produces wrong numbers, or behaves unexpectedly            |
| <b>Correction</b>  | Something is factually wrong — a method, equation, IPCC reference, or number |
| <b>Enhancement</b> | A feature or change that would make the tool more useful or easier to use    |
| <b>General</b>     | A general comment, question, or observation                                  |

| Severity        | Meaning                                                               |
|-----------------|-----------------------------------------------------------------------|
| <b>Critical</b> | Must fix before release — a calculation error or crash                |
| <b>High</b>     | Should fix before release — a significant accuracy or usability issue |
| <b>Medium</b>   | Can fix in the next iteration — a minor issue that doesn't block use  |
| <b>Low</b>      | Nice to have — cosmetic or very minor                                 |

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Severity
Meaning

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## Step 5 — Send it back

Save the completed `beta_feedback_template.xlsx` and **email it to [M.Lolita@cgiar.org](mailto:M.Lolita@cgiar.org)**. If several people are reviewing, each can send their own file — the *Reviewer name* column keeps everyone’s comments distinct.

## Two things you might notice (and can ignore)

- The green “**Feedback**” button at the bottom-right of the app is for quick, one-off bug reports once the tool is in wider use. For this structured review, please use the **Excel form** above instead.
- `questionnaire.md` / `questionnaire.docx` (in the Resources area) is a data-preparation form for the *AI Translator* — it is **not** this feedback form. You can ignore it for the review.

## What makes feedback most useful

- **Be specific** — name the tab and the exact button, table, chart, or text.
- **Say what you expected** versus what actually happened.
- **One observation per row**, so each item can be tracked and resolved individually.

Thank you — your review directly shapes what we fix before the tool is released more widely.